

Evaluate limits using L'hospital's rule

Key Points

L'hospital's rule states that

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

That is we can differentiate the numerator and denominator to help make the easier to compute

Technicalities:

- $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0$ or $\pm \infty$
- $g'(x) \neq 0$ for all $x \neq a$

Don't worry too much about these technicalities though

Basic Skills

Q1. Differentiate the following functions

a) $x^2 - 5x^4$ b) $\sqrt[3]{x+2} + 2$

c) $\sin^2 x$ d) $\ln^2(x)$

Q2. Given that $f(x) = x^2 \sin x$ and $g(x) = \ln x$

a) Show that $f'(x) = 2x \sin x + x^2 \cos x$ and $g'(x) = \frac{1}{x}$

b) Show that $\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = \lim_{x \rightarrow 0} x^2(2 \sin x + x \cos x)$

c) Hence state the value of $\lim_{x \rightarrow 0} \frac{f(x)}{g(x)}$

Q3. Use L'hospital's rule to evaluate $\lim_{x \rightarrow \infty} \frac{3+5x}{6x-5}$

Competency Skills

Q4. Is one application of L'hospital's rule always enough to evaluate the limit? Give a reason

Q5. For the limit $\lim_{x \rightarrow 1} \frac{4x^2 - 7x + 3}{\cos(\frac{\pi x}{2})}$

a) State the current problem with the limit?

b) Use L'hospital's rule to find the limit

Q6. Use L'hospital's rule to find the following limits

a) $\lim_{x \rightarrow 0} \frac{e^{2x} - x}{e^x - 1}$

b) $\lim_{x \rightarrow \infty} \frac{e^{3x}}{x^2}$

c) $\lim_{x \rightarrow -\infty} \frac{e^x}{x^3 + 2x}$

d) $\lim_{x \rightarrow \infty} \frac{x^3 - x + 5}{x^2 - 6x + 1}$

e) $\lim_{x \rightarrow 0} \frac{(6-5x^2) \sin x}{x \cos x}$

f) $\lim_{x \rightarrow \infty} \frac{x^2}{e^{x^2}}$

Q7. Use L'hospital's to help explain why the following limit doesn't exist $\lim_{x \rightarrow \infty} \frac{x^2 + \sin x}{5x^2 - 3}$

Advanced Skills

Q8. Explain why L'hospital's rule is the same as division when both $f(x), g(x)$ are polynomials in the limit $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)}$. Discuss 3 cases: $f(x)$ and $g(x)$ are the same order polynomial, $f(x)$ is a higher order polynomial than $g(x)$ and $g(x)$ is a higher order polynomial than $f(x)$.

Q9. For the limit $\lim_{x \rightarrow 0} \frac{\ln(\cos 2x)}{e^x - 1 - x}$

a) Evaluate it using L'hospital's rule

b) Evaluate it using Maclaurin's series

c) Compare the use of L'hospital's rule to Maclaurin's series in evaluating this limit. Does this apply to all limits?

Extension

Q10. What is the problem with using L'hospital's rule to work out $\lim_{x \rightarrow \infty} \frac{e^x - e^{-x}}{e^x + e^{-x}}$ (limit of hyperbolic tan). Can you come up with a remedy the problem and hence find the limit using L'hospital's rule

Hint: Multiply the top and bottom by something to remove the issue